

Crypto Price Prediction: Extra Credit Final Report

1. Introduction

This report documents the implementation and findings of the extra-credit extensions proposed for the crypto price prediction project. Each extension was directly motivated by a specific, measurable failure observed in the v9 baseline system. The goal was to understand *why* the baseline models failed and to apply targeted improvements that addressed each root cause.

Three extensions were proposed and implemented:

- Extension 1: Huber Loss + Attention LSTM, targeting LSTM mean collapse and non-local lag blindness
- Extension 2: XGBoost + Technical Indicators, targeting Ridge's linearity ceiling and interaction blindness
- Extension 3: Temporal Fusion Transformer (TFT), serving as a full architectural replacement for sequence modeling

All extensions were evaluated on the same held-out test set (Dec 2024 to Mar 2026, 433 days) using directional accuracy, Sharpe ratio, and P&L as primary metrics.

2. Baseline Project Results

The baseline project was trained on ETH-USD daily data from 2017 to 2026, using a 70/15/15 train/val/test split. Features included 30 return lags, log volume, and 14-day rolling volatility (32 features total). The two primary models, Ridge Regression and LSTM, were evaluated against two baselines on the held-out test period.

Model	Dir. Acc	MAE (return)	Sharpe	P&L
Last Value (baseline)	0.0%	N/A	N/A	N/A
7-day MA (baseline)	51.6%	N/A	N/A	N/A
Ridge Regression	54.2%	0.02850	1.14	+120.6%
LSTM	46.3%	0.02804	1.50 (neg)	159.3% (neg)

Key findings from the baseline ablation and interpretability analysis:

- Adding log volume alone raised Ridge directional accuracy from 53.0% to 55.3% (+2.3 pp), the single largest feature boost.
- Adding volatility_14 alone hurt accuracy (52.97% to 52.26%), but combining it with volume helped, revealing a conditional interaction effect Ridge cannot capture.
- Permutation importance identified lag_9, lag_16, and lag_30 as top features, not recent lags like lag_1 or lag_2.
- Residuals clustered during high-volatility regimes, where the model consistently under-predicted sharp moves.
- The LSTM made only 61 trades vs. Ridge's 169, with its learning rate reduced 6+ times before early stopping, a clear sign of mean collapse.

3. Problems Addressed

The three extensions each responded to a specific, documented failure from the baseline results.

Problem 1: LSTM Collapsed to Near-Zero Predictions

The LSTM's 46.3% directional accuracy (below random) combined with only 61 trades revealed the model converged to predicting the mean return (approximately 0) rather than learning directional signal. This is a known consequence of training with MSE loss on heavy-tailed return data: the model learns to predict zero to avoid quadratic penalties on spike days, producing low MAE but zero directional utility.

Problem 2: Ridge Cannot Model Feature Interactions

Ridge's best regularization alpha was 99.27, an unusually high value forcing near-mean predictions. More critically, the ablation revealed a conditional interaction effect Ridge cannot capture: volatility alone hurt accuracy, but combined with volume it helped. This suggests the signal is conditional on market context, which is structurally invisible to a linear model.

Problem 3: Non-Local Lags Dominate, but LSTM Processes Sequentially

The top three permutation importance features were lag_9, lag_16, and lag_30, not the most recent lags. The LSTM compresses all 30 timesteps through a sequential hidden state, which structurally buries signal from specific, distant timesteps. A model able to directly attend to non-adjacent lags would be better suited to this pattern.

4. Extensions Implemented

All three extensions were implemented and integrated into the proposal-complete notebook with `SAFE_MODE` execution controls for reliable Colab runs.

Extension 1: Huber Loss + Attention LSTM

Addresses: Problem 1 (mean collapse) and Problem 3 (non-local lags)

The MSE-based LSTM was replaced with a Huber loss formulation ($\delta \approx 0.03$, approximately one standard deviation of ETH daily returns). Huber loss behaves like MSE for small errors but linearly beyond δ , removing the disproportionate quadratic pull toward zero on spike days. A single-layer additive attention mechanism was added after the second LSTM layer, allowing the model to directly weight any of the 30 lookback timesteps rather than relying solely on the final hidden state.

- Huber loss removes the quadratic penalty that causes mean collapse during high-volatility periods.
- Attention allows the model to learn that lag_9 and lag_30 matter more than lag_1, matching the permutation importance finding.
- Optuna tuned δ alongside existing hyperparameters; the LSTM architecture was preserved with minimal code change.

Extension 2: XGBoost with Technical Indicators

Addresses: Problem 2 (linearity ceiling and feature interactions)

Ridge was replaced with an XGBoost regressor, which captures nonlinear interactions natively through tree splits. The feature set was extended with three technical indicators computable from existing OHLCV data: RSI(14), MACD signal line, and Bollinger Band width. An interaction feature *vol_x_volume* was also added. A 5-group ablation was run (matching the previous project's 4-group structure with a fifth group for the new indicators), along with regime-based comparisons across high-volatility and high-volatility-high-volume market subsets.

- XGBoost can model high volume and high volatility as a combined regime signal, capturing the conditional effect Ridge missed.
- RSI(14) captures momentum saturation absent from raw return lags. MACD captures trend divergence at two timescales.
- Bollinger Band width normalizes volatility more cleanly than raw rolling standard deviation.

- The existing data pipeline required minimal changes; all new features derive from existing OHLCV columns.

Extension 3: Temporal Fusion Transformer (TFT)

Addresses: Problems 1 and 3 at the architectural level

The LSTM was replaced entirely with a Temporal Fusion Transformer, a sequence model combining multi-head self-attention, gating mechanisms, and a variable selection network. TFT directly connects any two timesteps via attention, meaning lag_30 is as accessible as lag_1 without passing through 29 intermediate hidden states. Its variable selection network provides per-timestep feature gating analogous to the permutation importance results from the baseline project.

- Multi-head attention solves the non-local lag problem structurally, not as a patch.
- Variable selection network provides interpretable attention weights comparable to permutation importance.
- Implemented via pytorch-forecasting's TFT module; Optuna tuned hidden_size, attention_head_size, and dropout.
- A try/except skip path was included for computational stability; TFT is activated via the RUN_TFT=True flag.

Engineering and Execution Support

The notebook was built progressively across four versions (v2, v3, v3 stable, proposal-complete) to balance full coverage with Colab reliability. Additional engineering work included:

- SAFE_MODE flag: when set to True, skips TFT and heavy Optuna passes by default for a reliable Run All
- Validation threshold tuning with a minimum-trade guard and cost sensitivity analysis at 5, 10, 15, and 20 bps
- Section 17/18 graphical diagnostics including Sharpe/P&L bars, cumulative curves, cost lines, and threshold sweep
- Section 18 proposal compliance checklist (PASS/FAIL table), final score 7/8 PASS

5. Results

All models were evaluated on the held-out test set using the same metrics as the baseline project. Run mode: *SAFE_MODE=False*, *RUN_TFT=True*, *RUN_LSTM_OPTUNA=True*, all three extensions active. Compliance checklist score: **7/8 PASS**. The single FAIL item was XGBoost Sharpe (0.82) falling below Ridge Sharpe (1.12).

Final Full-Coverage Run

Model	Dir. Acc	Sharpe	P&L	Notes
Ridge v9 baseline	52.66%	1.1194	+1.1403	Best Sharpe/P&L
XGBoost (raw)	53.12%	0.3839	+0.3922	Best Dir. Acc
XGBoost (tuned)	53.12%	0.8185	+0.6181	Strong after tuning
Huber + Attn LSTM	50.12%	-0.0664	-0.0679	Negative this run
TFT	52.19%	0.3693	+0.3777	Positive, secondary
LSTM v9 baseline	51.73%	0.6752	+0.6898	Strong in this run

Notable in this run: the LSTM v9 baseline performed strongly (Sharpe 0.6752, P&L +0.6898), which is a reversal from its collapsed behavior in earlier runs. This highlights how sensitive model rankings can be to training randomness and threshold calibration across runs.

6. Findings by Topic

LSTM Stabilization via Huber + Attention (Extension 1)

Finding: Not successful in this run.

Huber + Attention LSTM posted negative Sharpe (-0.0664) and negative P&L (-0.0679) in the final full-coverage run, placing it last among all models on trading metrics. It also did not outperform the baseline LSTM or Ridge on directional accuracy (50.12% vs 51.73% and 52.66% respectively). The architectural intent was sound, but the extension was not robust under the current runtime and training configuration.

This is a more unfavorable result than earlier runs suggested. The mean collapse problem was targeted correctly at the design level, but optimization sensitivity, training budget constraints, and threshold policy fragility appear to have prevented the gains from materializing in this particular run.

Nonlinear Interactions via XGBoost (Extension 2)

Finding: Directional improvement confirmed; trading improvement partial.

XGBoost variants achieved the highest directional accuracy in this run at 53.12%, slightly above Ridge at 52.66%. This continues to support the proposal claim that nonlinear models can improve directional signal capture. However, directional improvement did not translate into the best Sharpe or P&L in this run: Ridge led both at 1.1194 and +1.1403 respectively. XGBoost tuned showed notably stronger trading performance than XGBoost raw (Sharpe 0.8185 vs 0.3839), confirming that threshold policy and execution tuning matter significantly for this model class.

Non-Local Temporal Modeling via TFT (Extension 3)

Finding: Partial viability confirmed.

TFT executed successfully and produced positive Sharpe (0.3693) and P&L (+0.3777) in the final run, confirming it is a workable component in the pipeline. It did not become a top performer relative to Ridge or XGBoost tuned under the current CPU-constrained setup. This result is consistent with the proposal's framing of TFT as an architectural benchmark: it demonstrates the mechanism works, but realizing its full potential would require additional training budget and hyperparameter tuning depth.

Objective Mismatch Between Direction and Trading Quality

Finding: Strongly confirmed again.

The most consistent finding across all runs remains that the best directional model and the best trading model are not the same. In this run, XGBoost led directional accuracy while Ridge led Sharpe and P&L. The gap between XGBoost raw (Sharpe 0.3839) and XGBoost tuned (Sharpe 0.8185) also reinforces that threshold policy calibration has a larger effect on trading outcomes than model architecture alone. Model selection should prioritize trading utility, meaning Sharpe/P&L under realistic transaction costs, rather than directional accuracy.

7. Key Discoveries

#	Discovery
1	Directional accuracy and trading performance diverge consistently. The best directional model and the best trading model were not the same in any run.
2	XGBoost provided the best directional accuracy in this run (53.12%), supporting the nonlinear interaction hypothesis from the proposal.
3	Ridge remained the strongest trading benchmark with Sharpe 1.1194 and P&L +1.1403, despite not leading on directional accuracy.
4	Huber + Attention LSTM was not robust in this run: it produced negative Sharpe and P&L, placing last on trading metrics. This does not invalidate the architectural approach but does reflect sensitivity to training configuration.
5	TFT is workable but secondary: it produced positive results but did not compete with Ridge or XGBoost tuned under current compute constraints.
6	Threshold tuning materially changes rankings. XGBoost tuned (Sharpe 0.8185) substantially outperformed XGBoost raw (Sharpe 0.3839), showing that policy calibration matters as much as model choice.

8. Conclusion

All three proposed extensions were implemented and executed in the final notebook, with 7/8 compliance items passing. The single failing item was XGBoost Sharpe falling below Ridge Sharpe, reinforcing the objective mismatch finding. **Ridge remains the best trading model** (Sharpe 1.1194, P&L +1.1403), while **XGBoost remains direction-competitive** and interpretable for nonlinear effects (53.12% directional accuracy). Sequence model extensions did not become top performers under the current setup: Huber + Attention LSTM posted negative results in this run, and TFT produced positive but secondary results.

The main methodological lesson remains **objective alignment**: in trading tasks, final model selection should be based on Sharpe/P&L under realistic costs, not directional accuracy alone. The gap between XGBoost raw and XGBoost tuned further shows that execution policy calibration can matter as much as the model architecture itself.

Recommended Next Steps

If this work were to continue, the following directions are most justified by the evidence:

1. **Select the final model by trading objective.** Use Sharpe/P&L under realistic transaction costs as the primary selection criterion, not directional accuracy.
2. **Re-run threshold tuning with stricter anti-overfit controls.** Rolling validation or nested validation windows would reduce the policy overfitting risk; the XGBoost raw vs tuned gap illustrates how much tuning matters.
3. **Re-evaluate Huber + Attention LSTM with additional training budget.** The negative result in this run likely reflects optimization sensitivity rather than a fundamental flaw in the architecture.
4. **Keep TFT as optional only.** Do not invest further in TFT unless additional training budget and hyperparameter tuning depth are available, as incremental gains do not yet justify the compute cost.